

COYOTE VOCALIZATIONS: A LEXICON AND COMPARISONS WITH OTHER CANIDS

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Abstract. Vocalizations of free-ranging and captive coyotes (*Canis latrans*) were recorded from January 1972 through August 1975. A lexicon of 11 vocalizations was constructed based on their: (1) sounds as perceived by the observer, (2) behavioural context, and (3) physical characteristics as determined through sound spectrographic analysis. The suspected functions and relative intensities of communication of the vocalizations are described. Comparisons are made with the vocalizations of other canids.

Coyotes (*Canis latrans*) are the most vocal of all North American wild mammals (Gier 1975), but their vocalizations have seldom been studied. The most extensive study has been McCarley's (1975) spectrographic analysis and description of long-distance vocalizations. Vocalizations have been mentioned as an adjunct to more general research on coyotes (e.g. Murie 1940; Brown 1973) and in reviews of mammalian vocalizations (Tembrock 1959b, 1963, 1968).

This paper presents a lexicon of coyote vocalizations based on direct observation and spectrographic analysis. The lexicon is conservative and includes only those vocalizations which are easily differentiated perceptually by humans, and which I suspect communicate different information. I have excluded vocalizations which may represent only intra-individual variation, although they may have different communicative value, and I have opted for the simplest lexicon until further pragmatic (Marler 1961) research demonstrates the need to split the vocalizations further.

This investigation utilized the syntactical approach which '... is concerned primarily with describing and classifying signals and attempting to extrapolate to predictions about the potential properties of the signal system when used in actual communication' (Marler 1965, p. 546).

Methods

Subjects and Study Areas

Both captive and wild coyotes were studied from January 1972 through August 1975. Coyotes of both sexes and all age classes (puppies to adults) were observed, and vocalizations from all groups were recorded and analysed. These included 'spontaneous' vocalizations of all types, as well as long-distance

vocalizations elicited by playback and human howling.

Captive coyotes at the Colorado State University's Maxwell Ranch were observed periodically throughout the entire study. These included both sexes, and ages from 4 months to over 8 years. They were kept in 10 kennels in groups of one to three. Observations were also made while the coyotes were in a 6400-m² enclosure in conjunction with other research.

Observations of free-ranging coyotes were made in Grand Teton National Park and on the National Elk Refuge, Jackson, Wyoming, for 5 and 6 weeks during the summers of 1974 and 1975, respectively. Also, occasional observations were made in the Fort Collins area throughout the study.

Methods of Recording and Analysis

Vocalizations were recorded with a Nagra IV-L audio tape recorder, an Electro-Voice 654A microphone and a Tornøgren 61-cm parabolic reflector. The parabolic was either hand-held or mounted on a tripod. Recordings were made on BASF DP 26 magnetic tape at 19 cm/s. Approximately 30 h of recordings were made from both free-ranging and captive coyotes.

The vocalizations were initially categorized according to the differences in sound that I perceived while hearing them. Sound spectrographic analysis was then used to check for consistency within categories and to provide a physical description for each. A total of 790 sonagrams was produced using a Kay Sona-Graph model 7029A with the FL-1 filter. The wide band pass filter was used for both display purposes (Fig. 1) and temporal measurements, and the narrow band pass filter was generally used for frequency (Hz) measurements.

Results and Discussion

The vocalizations were segregated into 11 categories based on their different sounds, behavioural context, and physical characteristics. The order of the vocalizations in Fig. 1 represents a physical continuum (graded system) and intermediates can be demonstrated for most of the adjacent vocalizations. The vocalizations, the range of their dominant frequencies (Marler

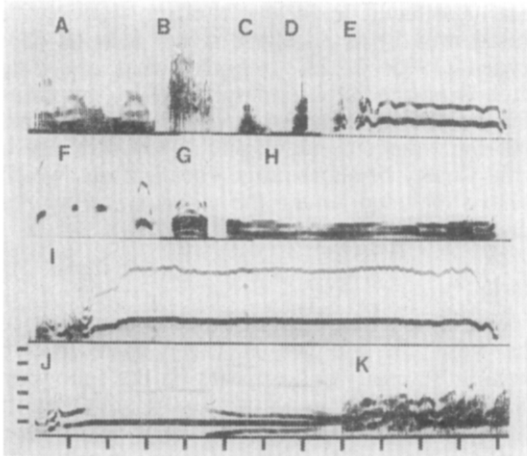


Fig. 1. Sound spectrograms of the eleven coyote vocalizations. A, growl; B, huff; C, woof; D, bark; E, bark-howl; F, yelp; G, whine; H, wow-oo-wow; I, lone howl; J, group howl; K, group yip-howl. The vertical scale is marked in 1-kHz intervals. The horizontal scale is marked in 0.25-s intervals.

1973), and a description of their physical and phonetic characteristics are listed in Table I. The functions suggested for each vocalization were deduced from observations of the transmitting and receiving coyotes, and the general behavioural context in which the vocalizations occurred (Table II).

Growl

The growl is a 'gr-r-r-r' sound used as a short-range threat in an agonistic context. It is generally a rather low intensity threat but can vary in conjunction with the positioning of the lips. It is of low intensity when the 'gr-r-r-r' is of low amplitude and the lips are covering all but the tips of the canines. It can develop into a high intensity threat display with a high amplitude 'gr-r-r-r' and the lips pulled back to expose the canines and incisors (Fox 1970). This latter display has been referred to as a 'snarl' (Bekoff 1972; Brown 1973; Murie 1940; Pringle 1960; Silver & Silver 1969).

Coyote pups utilize growling in fighting and play fighting at about 18 days of age (Fox 1969; Bekoff 1972). In all canids it appears to serve as a threat in an agonistic encounter, or in a play-fighting context in pups. In addition, Bekoff & Diamond (1976) reported that low growls were given by a male coyote during precopulatory behaviour.

Huff

The huff is another short-range, low intensity threat that is produced by rapid expulsion of

Table I. Coyote Vocalizations Listed in Order Along a Physical Continuum

Vocalization	Range of dominant frequency* (kHz)	Description
Growl	0.3-1.6	'Gr-r-r-r' sound; low-medium amplitude.
Huff	0.5-3.0	Rapid expulsion of air through mouth and nose.
Woof	0.2-0.7	Low amplitude bark.
Bark	0.6-2.1	Burst of high amplitude and short duration; appear as wide frequency range noise or concentration of energy into narrow frequency bands.
Bark-howl	0.3-3.0	Frequency modulated short howl; interspersed with a series of barks and generally immediately preceded by a bark; high amplitude.
Whine	0.3-3.0	Spectrographically resembles a low frequency howl with several harmonics; low amplitude.
Wow-oo-wow	0.3-1.3	Howl of low amplitude and frequency which modulates in both frequency and amplitude.
Yelp	0.7-2.5	'Yelp' portion resembles a high frequency bark: 'yi-e-e-e' portion resembles a short howl broken into segments.
Lone howl	0.4-2.0	High amplitude; slight-moderate frequency modulation; relatively long duration.
Group howl	0.4-2.3	Same as lone howl but given by two or more coyotes polyphonically or antiphonally; generally with different dominant frequencies.
Group yip-howl	0.3-3.0	High amplitude; rapid yipping sounds interspersed with howls of generally shorter duration than the lone howl or group howl.

*Midpoint of the blackest part of the spectrogram of the vocalization (Marler 1973).

air through the mouth primarily, but also the nose. Phlegm is often expelled through the nose during a huff and is sometimes followed by a sneeze. I suspect that the 'sneezing' reported for golden jackals (*Canis aureus*) by Seitz (1959) is essentially the same vocalization that I am calling a huff. Seitz (1959, p. 770) says that the 'sneeze' is a 'repeated audible exhalation' which 'with increasing disturbance joint barking occurs'. In contrast, I know of no instance where a huff was given as an alarm vocalization. Eaton (1969, p. 88) described an observation of an adult black-backed jackal (*Canis mesomelas*) making '... a very pneumatic sound by forcing air out of the lungs in short staccato bursts' which was directed towards cheetahs. The red fox (*Vulpes vulpes*) is reported to give a 'nasal coughing note' when angry (Huxley & Koch 1938). Bleicher (1963) reported that the domestic dog (*Canis familiaris*) makes a puffing sound by forcing air through the slightly opened mouth and nose, during the lowest intensity of close vocal aggressive display. A '... chuffing sound like a series of rapid exhalations ...' was reported by Silver & Silver (1969, p. 25) for their captive New England canids (Eastern coyotes) (*Canis latrans* var.). They stated that it served as a warning that something out of the ordinary was taking place and for want of a better term they called it barking. This vocalization may be what I have called a huff, or since it was used in an alarm context it may be what I have called a woof. Brown (1973, p. 151) and Bekoff (personal communication) reported that their captive coyotes gave a vocalization similar

to a woof which was '... simply a very rapid expiration'.

Woof

The woof is a low amplitude bark which apparently serves two functions: (1) low intensity threat, and (2) low intensity alarm. Pups respond to a woof by returning to the den and generally going below ground (Camenzind 1974). Tembrock (1963) indicated that the red fox uses a faint, difficult to locate, barking sound (with closed mouth) as an alarm signal near its den. Ryden (1975, p. 227) reported that a coyote bitch sent the pups underground '... by emitting a soft vocal signal that sounded like a short half-wail, descending in pitch'. Coyotes use a wide variety of short-range vocalizations which I have included under the category of whines (discussed below).

Bark

The bark is the long-distance counterpart of the woof. In addition, it varies from low to medium intensity and appears on the sonogram as a burst of high amplitude and relatively short duration (Fig. 1D). The frequency distribution can vary from wide frequency noise to relatively discrete bands of the dominant frequency plus harmonics (Figs. 2 and 3) (McCarley 1975). Fox (1971) stated that coyotes emit barks in three behavioural contexts: (1) when alarmed, (2) in territorial defence, or (3) for the co-ordination of hunting movements. I concur with the first two, but I know of no evidence to support the third

Table II. Coyote Vocalizations and the Behavioural Context in Which They Occurred

Vocalization	Behavioural context	Distance	Intensity of communication*
Woof	Agonistic (threat); alarm	Short	Low
Growl	Agonistic (threat)	Short	Low-High
Huff	Agonistic (threat)	Short	High
Bark	Agonistic (threat); alarm	Long	Low-Med.
Bark-howl	Agonistic (threat); alarm	Long	High
Yelp	Agonistic (submissive); startle	Short	High
Whine (high frequency)	Agonistic (submissive)	Short	Low
Whine (low frequency)	Greeting	Short	Low
Wow-oo-wow	Greeting	Short	High
Lone howl	Separated group members; response to distant howling	Long	Low-High
Group howl	Separated group members; response to distant howling	Long	Low-Med.
Group yip-howl	Reunion of group members; response to distant howling	Long	High

*See Discussion section for explanation.

in coyotes. However, Eaton (1969) reported that the black-backed jackal uses barking to maintain contact while they move and also to co-ordinate hunting activities. The bark has been referred to as an alarm vocalization in the golden jackal (Seitz 1959), grey wolf (*Canis lupus*) (Joslin 1966; Scott 1967), domestic dog (Scott 1968; Scott & DeGhett 1972), African hunting dog (*Lycaon pictus*) (Estes & Goddard 1967; Schaller 1973), red fox (Ables 1975), and the bush dog (*Speothos venaticus*) (Bates 1944). It is referred to as a threat vocalization in the coyote (Young & Jackson 1961; Brown 1973; Camenzind 1974), golden jackal (Seitz 1959), grey wolf (Joslin 1966; Theberge & Pimlott 1969; Mech 1970), domestic dog (Darwin 1872; Bleicher 1963; Pulliainen 1967), African hunting dog (Kuhme 1965; Schaller 1973), and red fox (Burrows 1968). Barking apparently promotes spacing of individual maned wolves (*Chrysocyon brachyurus*) over long distances (Kleiman 1972), and may assist in setting up territories and announcing their occupancy in red foxes (Burrows 1968). Scott & DeGhett (1972) suggested that barking in the domestic dog may serve as a noxious stimulus to the listener.

There is disagreement over the relative ease with which the source of a bark can be localized.

Scott (1961; 1968) and Scott & DeGhett (1972) stated that the domestic dog's bark is easily localized, and Tembrock (1963) said the same for the red fox's bark. Ryden (1975) and Bekoff (personal communication) reported that they had difficulty locating the source of a coyote's bark. The physical characteristics of the bark (generally wide frequency range and short duration) suggest that the bark should be easily located, and I (also McCarley personal communication) have found this to be the case with coyotes. Kiley (1972) stated that a single dog bark is difficult to locate, but if it is repeated then it becomes easy to locate. Camenzind (1974) described the use of the 'muffled "woof" bark' near the coyote's den which causes the pups to take cover; then loud barking bouts are used as a warning and threat after the adults have moved several hundred yards from the den area. I have observed similar behaviour which indicates that coyotes do not bark repeatedly in close proximity to the den.

Burrows (1968) was able to distinguish the barking of individual red foxes by differences in pitch and in the number of barks forming a sequence. Domestic dog barks are reported to be produced in '... a highly similar series by any particular individual' (Scott 1961, p. 387). Tembrock (1965) suggested that the frequency, duration and repetition of canid barks may determine the contents of information, and that the amplitude and frequency range may also be determining features. I measured the frequency and duration of barks for six individual free-ranging coyotes. All of the barks were given in

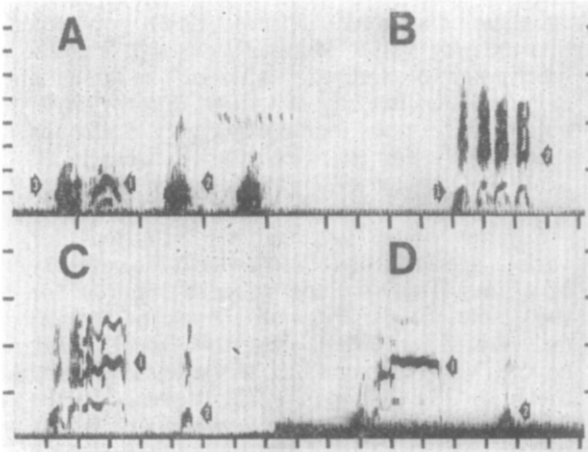


Fig. 2. A. Differences in characteristics of an individual coyote's barks from rather pure frequencies (1), to bursts of wide frequency 'noise' (2), and barks intermediate between the two (3). These are four successive barks from the same individual. B. Change in dominant frequency from a lower (1) to a higher (2) frequency in a series of barks. C and D. Change in dominant frequency from a bark-howl (1) to a bark (2). The vertical scale is marked in 1-kHz intervals. The horizontal scale is marked in 0.25-s intervals.

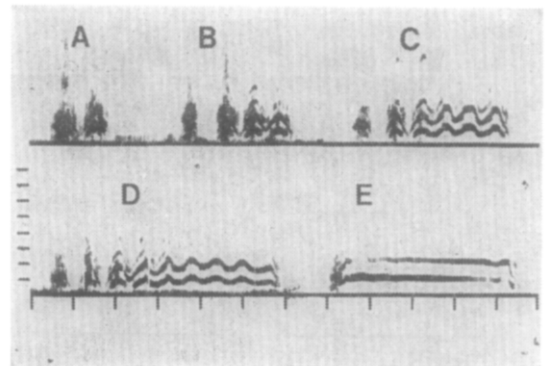


Fig. 3. Development of the bark-howl (C, D, E) from the bark (A, B). These were selected from a single bout by an individual coyote. The vertical scale is marked in 1-kHz intervals. The horizontal scale is marked in 0.25-s intervals.

response to my presence and all barks measured for each individual were taken from the same bout. Both parameters were too variable for significant differences between individuals (Fig. 4). This suggests that individual recognition, if encoded at all, is based on other characteristics such as formant shape (Theberge & Falls 1967; Kiley 1972). However, there was considerable variability in the frequency distribution within each individual's bout which varied from wide frequency noise to discrete bands (Fig. 2A). Also, the dominant frequency often changed in series of barks (e.g. Fig. 2B). I suggest that individual recognition is less important information than location and alarm in the long-distance bark, and that variation and gradation into the bark-howl may provide important information on the intensity of communication at the possible expense of individual recognition.

Bark-Howl

The bark-howl is a short, frequency modulated howl, interspersed with a series of barks and generally preceded by a bark (Fig. 1E). It appears to serve the functions of a high intensity bark, both alarm and threat. It develops from barks being drawn together and then integrated into a short howl (Fig. 3). This suggests that the bark-howl may have evolved from the bark as the coyote increased its intensity of communication along the continuum shown in Fig. 3. The bark-howl may have become ritualized and taken on signal value as a vocalization by itself. Indeed, it is often given as a distinct vocalization separated from barks by several seconds. McCarley (1975) has referred to the same

vocalization as a 'warble howl'. In addition to high intensity alarm and threat, the howl portion might make the source easier to locate (Kiley 1972), as well as carry additional information. The dominant frequency often changes as the vocalization changes from the bark to the howl portion and back again (Figs. 2C, 2D).

The bark-howl has been reported to serve as an alarm in coyotes (Alcorn 1946; Benson 1948), grey wolves (Stephenson & Ahgook 1975), and dingos (*Canis dingo*) (Corbett & Newsome 1975). Joslin (1966) reported an incident in which a grey wolf used a 'drawn-out bark' to threaten him.

Pure barking bouts are consistently shorter than mixed barking, bark-howling bouts. Mech (1970) has recorded one bark-howling bout in grey wolves of 2½ h, and I have recorded a 4-h bout from a coyote.

Whine

The whine is a short-range, low amplitude vocalization. Spectrographically it resembles a low frequency lone howl with several harmonics (Fig. 1G). In adult coyotes, a low frequency whine is given by a subordinate to a more dominant individual in a low intensity greeting (active submission), and high frequency whines are used by subordinate animals to express low intensity submission (passive submission), in an agonistic encounter. Joslin (1966) reported that the grey wolf's 'whimper' is used in friendly greeting or to express submission. The whine is et-epimeletic in coyote pups (Snow 1967), golden jackal cubs (Seitz 1959), grey wolf pups (Scott 1950), domestic dog pups (Bleicher 1963; Scott & DeGhett 1972), and raccoon dog pups (*Nyctereutes procyonoides*) (Seitz 1955).

Several other vocalizations reported for various species probably fall within the category of whines. These include 'squeaks' reported for the New England canid (Eastern coyote) (Silver & Silver 1969), grey wolf (Mech 1970), coyote (Van Wormer 1964; McCarley 1975), and the bush dog (Kleiman 1972); 'squeals' in the bush dog (Bates 1944); and 'whimpers' in the golden jackal (Van Lawick-Goodall & Van Lawick 1970), grey wolf (Joslin 1966), and the domestic dog (Fuller & Fox 1969). Tembrock (1963) has described 'whimpers' in the dhole (*Cuon alpinus*), African wild dog, Corsac fox (*Vulpes corsac*) and red fox.

There are numerous whine vocalizations which are given by coyotes in different contexts. Most of these are very soft and can be heard only at

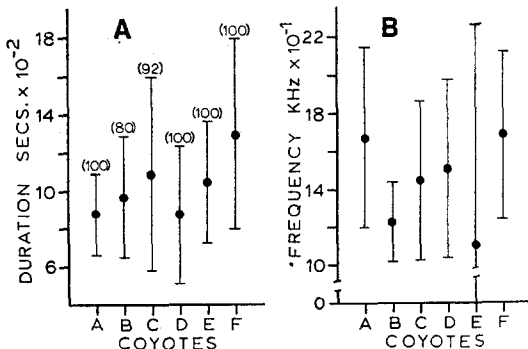


Fig. 4. A. Mean duration of barks from six individual coyotes. B. Mean dominant frequency of barks from the same six coyotes. The vertical lines represent 2 SE of the mean. The numbers in parentheses are the sample sizes.

close range, similar to the 'social squeaks' Mech (1970) has observed in grey wolves. Ryden (1975) and Camenzind (personal communication) reported that coyote pups are called from the den by adult females, probably by a vocalization, although it has not been heard by either observer. Female golden jackals call their cubs with a whine (Van Lawick-Goodall & Van Lawick 1970). Whines may also be given during mating and parturition, as reported for the raccoon dog (Seitz 1955) and the domestic dog (Bleicher 1963), respectively. However, as in the domestic dog (Fuller & Fox 1969), there do not appear to be any whine vocalizations universally associated with courtship or copulation in the coyote.

Wow-oo-wow

The wow-oo-wow is a short-range, low amplitude vocalization which modulates slowly both in frequency and amplitude (Fig. 1H). It appears to function as a high intensity greeting display and has been referred to elsewhere as a 'greeting song' (Camenzind & Lehner 1975). This is probably what Crisler (1958) has termed 'wowing' in grey wolves. It is generally given by a subordinate (juvenile or adult) while muzzlenibbling a dominant and showing a simultaneous combination of submissive approach and bite intention. The wow-oo-wow appears to be an ambivalent vocalization which occasionally grades back and forth between a growl (low intensity threat) and whine (low intensity submission). It is a mixed sound which apparently results from a blending of motivations (Tembrock 1968). Fentress (1967) reported that his captive grey wolf gave a 'growl-whine' which indicated a fluctuation between aggression and submission. Also Tembrock (in Kiley 1972, p. 217) suggested that the zebra's call of fluctuating pitch reflected a 'superimposition' of approach and withdrawal. Schenkel (1967) suggested that subordinate grey wolves may 'behave obtrusively' toward a dominant individual when the subdominant's active submission is met with tolerance and lack of display of superiority from the dominant individual. The wow-oo-wow vocalization is often given by domestic dogs to their keepers when reunited following an extended separation.

Yelp

The yelp is used in two behavioural contexts: (1) high intensity submission, and (2) startle or high intensity alarm (including pain). The yelp portion resembles a high frequency bark. It is

often followed (or preceded) immediately by a 'yi-e-e-e' which resembles a short howl broken into segments (Fig. 1F). McCarley (1975) recorded 'yipes' from elicited coyote 'choruses' which appear the same, spectrographically, as what I have designated as yelps. The yelp is probably similar to the 'cry of pain' which Seitz (1959) attributes to the golden jackal.

Lone Howl

The lone howl is the 'typical' long-distance vocalization attributed to the coyote. It is of high amplitude, has slight to moderate frequency modulation, and is of relatively long duration. The lone howl is essentially synonymous with McCarley's (1975) 'flat howl'. It is often preceded by a series of barks, referred to as 'herald barks' by Mengel (1971) for his captive coyote-dog hybrids. The lone howl is given in response to distant howling, as well as 'spontaneously' by separated group members. It is believed to aid separated group members in determining the location of each other and in some cases leads to a reunion. This has been suggested for the coyote (Ryden 1975), grey wolf (Jordan et al. 1967; Scott 1967; Theberge & Falls 1967; Theberge & Pimlott 1969; Mech 1970; Theberge 1971), black-backed jackal (Van Der Merwe 1953), golden jackal (Van Lawick-Goodall & Van Lawick 1970), and dingo (Corbett & Newsome 1975).

The lone howl, group howl and group yip-howl have been assigned separate status since they appear to serve different functions. Playback studies (Lehner 1976a, unpublished data) have demonstrated that coyotes respond differentially to group yip-howls versus lone howls or group howls. In addition, future research may determine whether there are different lone howls such as those attributed to grey wolves (Crisler 1958; Fentress 1967). Theberge (1971) reported that his three captive grey wolves could perhaps transmit emotional state by variations in the units of sound. I suspect that the gradations in coyote vocalizations are a reflection, in part, of emotional state; also, the large intra-individual and intra-howl variability demonstrates a capacity for coding large amounts of additional information (Fig. 5). It is the large intra-individual variation which would seemingly make it difficult to demonstrate significant inter-individual differences. However, Theberge & Falls (1967) were able to perceive differences between the howls of their three captive wolves and demonstrated spectrographically that

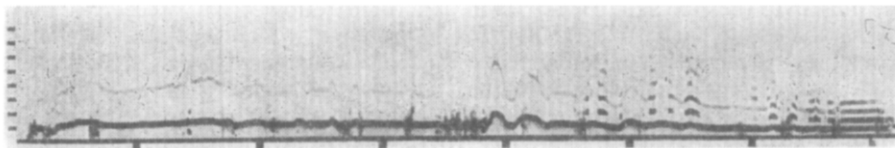


Fig. 5. Lone howl. The vertical scale is marked in 1-kHz intervals. The horizontal scale is marked in 1-s intervals.

differences existed in type of beginning, type of ending, pitch changes in the mid-section of howls, and significantly different average highest notes, lowest notes and lengths. In a study of wild wolves, Kolenosky & Johnston (1967, p. 302) found it '... almost impossible to distinguish individual wolves by their voices, except for a few animals with distinctive voices'. I can recall having heard only one lone howl by a coyote which I feel quite certain I could recognize if I heard it again.

Mech (1970, p. 102) states that 'Many people believe that howling is the call of the chase (in wolves) ... However, evidence for this is lacking'. Likewise, there is no evidence that coyotes howl during a chase. Cahalane (1961) observed three coyotes 'barking furiously' while chasing a deer, but Moran & Ozoga (1965) and Connolly et al. (1976) reported that no vocalizations were made by coyotes chasing elk calves and domestic sheep, respectively. Other observers have failed to report any vocalizations made by coyotes while chasing wild ungulates (Robinson & Cummings 1947; Bruns 1970).

Group Howl

The group howl is basically a combination of lone howls given by two or more individuals either antiphonally or polyphonically. When given polyphonically they avoid howling at the same dominant frequency. In Fig. 1J a second individual began howling midway in the sonagram and lowered its dominant frequency close to that of the first individual, at which point the first individual decreased its amplitude and stopped howling (off the sonagram). Ryden (1975) reported that she perceived the same phenomenon in the coyotes she was observing, and Crisler (1958) has reported it for grey wolves.

The group howl is sometimes given by two or more individuals separated from the rest of the group and is followed by a reunion. In addition, it may serve to announce territorial occupancy and reinforce group bonds (Camenzind 1974) but at a lower intensity than the group yip-howl.

The group howl also is apparently found in the golden jackal (Golani & Keller 1975), where it is reported to serve as a proclamation of territorial ownership (Van Lawick-Goodall & Van Lawick 1970).

Group Yip-Howl

The group yip-howl is a high amplitude, high intensity vocalization which includes several individuals howling and yipping together in seemingly distinct patterns (Fig. 1K). The howl portions are usually short and acutely frequency modulated. A group yip-howl bout is generally initiated by one individual, often the dominant individual (Brown 1973; A. Olsen personal communication), which is then joined by others (Fig. 6). The initiator generally gives a long, relatively un-modulated howl. As more individuals join in, the howls become more amplitude and frequency modulated and yipping (arrow in Fig. 6B) occurs, gradually becoming more frequent.

In addition to yips, McCarley (1975) also separated 'screams', 'gargles', and 'laughs' out of what I have collectively called a group yip-howl. McCarley also suggested that the 'scream' and 'gargle' may be juvenile variants of the 'flat howl'.

There are numerous references in the literature to unison howling in canids. Those vocalizations (terms) which appear to be synonymous (or at least very similar) to the group yip-howl in coyotes are shown in Table III. The function of the group yip-howl is probably to announce occupancy and defend a territory (or a large intergroup distance) and to reinforce group bonds (Camenzind 1974). It is believed to function in 'passive' territorial defence in coyotes (Tembrock 1959b; Camenzind 1974), jackals (Tembrock 1959b) and grey wolves (Joslin 1967; Mech 1970). Camenzind (1974) also observed its use in 'active' territorial defence during an inter-group conflict. In addition, it is sometimes given by coyotes

(Ryden 1975) and grey wolves (Crisler 1958) before setting off to hunt.

Elsewhere (Lehner 1976a, b), I have suggested that the group yip-howl may also be an 'epideictic display' and serve to regulate population density internally, a hypothesis first put forth by Wynne-Edwards (1962). Essentially the same hypothesis has been suggested for grey wolves (Joslin 1966, 1967). This could occur, not only through the spacing of groups and individuals, but also through direct reduction in reproduction. I suspect that the group yip-howl may provide individuals and groups with an assessment of coyote density, which when coupled with feedback from prey densities (Clark 1972) helps to regulate reproductive rate. Ryden (1975) and Pimlott (1960) suggest that they might be able to estimate the size of a group of howling coyotes and wolves, respectively, based on individual differences in the vocalizations; coyotes can

probably do equally well. Coyotes, like lions (Shortridge 1934, p. 82, quoting T. Roosevelt and E. Heller 1914), may vocalize less when their populations are low. This might be reflected in the increased fecundity shown by coyotes in artificially reduced populations (Knowlton 1972). Scent posts may provide redundant, but poorer and hence secondary, feedback into this system. My suggestion is that a high density of long-range, group yip-howl vocalizations may serve to reduce reproduction in coyotes in a manner similar to that shown for short-range olfactory communication (pheromones) in rodents (i.e. Bruce and Lee-Boot effects; Turner 1966). An initial group yip-howling bout might be metacommunicative, announcing the start of an 'assessment period', and the chain reaction of group yip-howling which then occurs in neighbouring groups would constitute the 'assessment period'. The larger and denser the groups, the more likely that an 'assessment period' will be initiated. It can be reasoned that it is advantageous to the individual and group (in terms of successfully raising young) to both transmit and receive in this system.

General Discussion

Graded Vocalizations

The lexicon of coyote vocalizations presented in this paper represents a graded continuum both within and between vocalization types. Consequently, the designation of vocalization categories was necessarily somewhat arbitrary and the assignment of recorded samples to particular categories was often subjective (e.g. bark versus bark-howl, Fig. 3B). This continuum has been described primarily on the basis of spectrographic characteristics, but there is strong evidence that gradations within and between vocalizations may also represent gradations in the quantitative (and perhaps qualitative) properties of the information being transmitted. I have expressed the quantitative differences as intensity of communication (Table II), a designation which is apparently synonymous with Kiley's (1972) term 'level of excitement'. As the intensity of communication increases, its relative credibility and urgency increases, and the level of response expected from the recipient increases. Also, if appropriate, the probability that the signaller will follow through increases (Smith 1969). For example, a growl is a threat which indicates concern and a desire for the recipient to leave the area. A higher intensity growl (higher amplitude and frequency) coupled

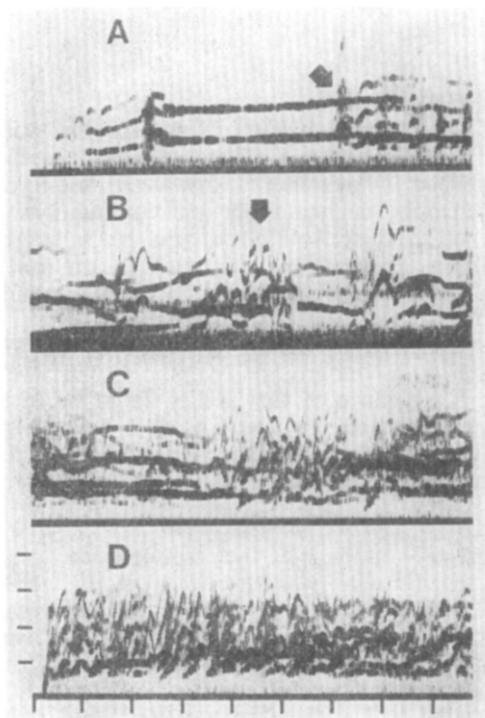


Fig. 6. Development of a group yip-howl bout. A. An individual coyote initiates the bout with a relatively unmodulated howl and is joined by a second coyote (arrow). As more individuals join (B, C, D) the more frequency and amplitude modulated the howls become and the more yipping (arrow in B) which occurs. The vertical scale is marked in 1-kHz intervals. The horizontal scale is marked in 0.25-s intervals.

Table III. Canid Vocalizations which are Believed to be Synonymous with the Group Yip-Howl. Numbers Refer to the Citations Below

Synonymous vocalizations	Species						
	Coyote	Wolf	Domestic dog	Dhole	Coyote-dog hybrid	New England wild canid (Eastern coyote)	New Guinea singing dog
Group howl	2, 10, 11						
Singing	1, 5	5				15	
Yapping chorus	14			4			
Evening song	13						
Happy social howl		3					
Chorus howl	14	12, 14	12		9, 15		
Communal howl	6	6					6
Irregular howl	8						
Howling concert							7
Combination howl	8						

1 Brown, J. B. 1973; 2 Camenzind, F. J. 1974; 3 Crisler, L. 1958; 4 Davidar, E. R. C. 1975; 5 Fox, M. W. 1971; 6 Kleiman, D. 1967; 7 MacIntosh, N. W. G. 1975; 8 McCarley, H. 1975; 9 Mengel, R. M. 1971; 10 Murie, A. 1940; 11 Ryden, H. 1975; 12 Scott, J. P. & V. J. DeGhett 1972; 13 Seton, E. T. 1909; 14 Seton, E. T. 1929; 15 Silver, H. & Silver, W. T. 1969.

with bared teeth (often referred to as a 'snarl') transmits that, with little doubt, if the recipient does not quickly retreat the transmitter will attack. Likewise, the huff (a different vocalization type) transmits essentially the same message as does the high intensity growl. In summary, intensity of communication can be altered by either a quantitative change (amplitude or frequency) or a qualitative change (vocalization type). Ewer (1968) concludes that the vocalizations of all species, other than humans, carry information only about the mood of the sender, and that none are capable of freeing the message from the mood. I am suggesting this same link for the coyote's vocalizations with the addition that they can vary the mood (intensity of communication) associated with the same message.

Gradation is probably a characteristic of canid vocal repertoires in general (Tembrock 1963), although Schaller (1972) considers the vocalizations of African wild dogs to be 'largely discrete', and Fox (1975) states that red fox vocalizations are 'stereotyped.'

Graded short and long-distance vocalizations are intuitively of adaptive significance for the coyote, a moderately social species which couples short-distance vocalizations with visual displays, and also relies heavily on long-distance vocal communication during nocturnal and solitary forays.

Ultrasonic Sounds

Circumstantial evidence suggests the possibility that coyotes produce ultrasonic sound, an aspect which needs investigation. Ryden (1975) and Camenzind (personal communication) report that the coyote bitch 'calls' her pups from the den, apparently by a vocalization that is either of too low intensity or too high frequency to be audible to observers near the den. Coyotes are receptive to sound as high as 30 kHz (Ewer 1973, based on Peterson et al. 1969). The significance of this ability could be in the detection of small prey (e.g. mice beneath the snow) and/or in intraspecific vocal communication.

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